

SURUTHI VIJAYA R

Full-Stack Developer | Java Backend Engineer | CSE (AI & ML) Graduate
+91 74188 42688 | suruthi.vijaya.aiml@gmail.com | [LinkedIn](#) | [GitHub](#) | Madurai, Tamil Nadu, India

CAREER OBJECTIVE

Recent B.E. graduate in Computer Science (AI & ML) with hands-on experience in Java, backend engineering, REST API design, and AI/ML systems. Seeking an entry-level Software Development Engineer or Backend Developer role to apply problem-solving skills and build scalable, clean-code solutions

EDUCATION

B.E. Computer Science — Artificial Intelligence & Machine Learning 2022 – 2026

Sethu Institute of Technology, Virudhunagar | CGPA: 8.47 / 10

Higher Secondary Certificate (HSC) — State Board 2022

OCPM Girls Higher Secondary School | Percentage: 83%

Secondary School Leaving Certificate (SSLC) — State Board 2020

OCPM Girls Higher Secondary School | Percentage: 78%

TECHNICAL SKILLS

Languages	Java (Proficient), SQL, Python
Backend Engineering	Node.js, Express.js, REST APIs, JDBC
Frontend & UI	React.js, HTML5, CSS3
Databases	MongoDB, MySQL
Tools & Platforms	Git, GitHub, Postman, VS Code, IntelliJ IDEA
Core CS Foundations	OOP, Data Structures & Algorithms (DSA)

WORK EXPERIENCE

Software Development Intern — InnoAivators (Remote)

- Collaborated with teammates on full-stack development of a cross-platform booking application, contributing to UI design, wireframing, and finalization.
- Built backend service architecture, establishing foundations for core API implementation and database schema deployment.
- Participated in sprint planning, technical scoping, and system design meetings under production-level constraints.

PROJECTS

Hospital Management System Next.js · Node.js · MongoDB · JWT · RBAC

- Engineered a full-stack web application structure automating workflows across 4 distinct healthcare roles: Doctors, Receptionists, Lab Technicians, and Pharmacists.
- Implemented secure token-based authentication and route protection using JSON Web Tokens (JWT) and Role-Based Access Control (RBAC) to enforce per-role data boundaries.
- Structured indexed MongoDB schemas optimised for Electronic Health Record (EHR) data organisation, retrieval efficiency, and role-scoped query patterns.

Student Management System

Java · JDBC · MySQL

- Developed a local backend CRUD application implementing the Data Access Object (DAO) pattern for clean separation of business logic and data-access concerns.
- Authored structured SQL relational queries — including SELECT, DELETE, and ALTER statements — to manage and maintain underlying student datasets.
- Applied OOP design principles, exception-handling routines, and clean code conventions as a practical bridge toward enterprise Java (Spring Boot) development patterns.

Lung Disease Analyzer

Python · FastAPI · Next.js · TensorFlow | Team of 2

- Collaborated in a 2-member team to architect a full-stack AI diagnostic system capable of identifying 4 lung disease from respiratory audio recordings.
- Spearheaded backend engineering by constructing an audio-to-spectrogram preprocessing pipeline and exposing model inference endpoints via FastAPI with structured API contracts.
- Integrated backend inference services with a Next.js clinical interface designed for usability, achieving 88.7% classification accuracy on the benchmark ICBHI dataset.

ACHIEVEMENTS

- Solved 100+ LeetCode problems in Java, focusing on data structures, algorithms, and OOP.
- Earned Gold Badge in Java and Bronze Badges in Python and SQL on HackerRank; completed certified assessments in both languages.

AREAS OF INTEREST

- Backend System Design & REST API Development
- Full-Stack Web Application Development
- Data Structures & Competitive Programming
- Artificial Intelligence & Machine Learning
- Database Design & Optimisation